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Lem diferenciacion lebesgue l1 imp loc l1

Lema 1. *Sea* $f \in \mathcal{L}^1_{loc}(\mathbb{R})$*. Supongamos que*

$$\forall g \in \mathcal{L}^1(\mathbb{R}) : \lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} g(t)\,\mathrm{d}t = g(x) \ \text{c.t.p. } x \in \mathbb{R}.$$

Entonces, $\lim_{h \to 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = f(x)$ *c.t.p.* $x \in \mathbb{R}$.

***Demostración*:** Definimos $\forall k \in \mathbb{N} : f_k = f \cdot \mathbb{1}_{[-k,k]}$. Como $f \in \mathcal{L}^1_{\mathrm{loc}}(\mathbb{R})$, sabemos que $\forall k \in \mathbb{N} : f_k \in \mathcal{L}^1(\mathbb{R})$. Por tanto, por hipótesis, tenemos que

$$\forall k \in \mathbb{N} : \exists E_k \subset \mathbb{R} : \left[m\left({E_k}^{\mathsf{c}}\right) = 0 \wedge \forall x \in E_k : \lim_{h \to 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f_k(x)\,\mathrm{d}x = f_k(x) \right].$$

Tomamos $E = \bigcap_{k=1}^{\infty} E_k$, entonces $m\left(E^{\mathsf{c}}\right) = m\left(\bigcup_{k=1}^{\infty} {E_k}^{\mathsf{c}} \right) \leqslant \sum_{k=1}^{\infty} m({E_k}^{\mathsf{c}}) = 0$.

Sea $x \in E$, entonces $\exists k_0 \in \mathbb{N}$ tal que $x \in [-k_0, k_0]$. Entonces, si tomamos $h_0 = k_0 - |x| > 0$,

$$\forall h \in (0, h_0) : [x-h, x+h] \subset (-k_0, k_0) \implies \forall t \in [x-h, x+h] : \mathbb{1}_{[-k_0,k_0]}(t) = 1.$$

Por tanto, para todo $h \in (0, h_0)$, tenemos que $\forall t \in [x-h, x+h] : f(t) = f_{k_0}(t)$, luego

$$\frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = \frac{1}{2h} \int_{x-h}^{x+h} f_{k_0}(t)\,\mathrm{d}t.$$

Tomando límite cuando $h \to 0^+$, obtenemos

$$\lim_{h \to 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f(t)\,\mathrm{d}t = \lim_{h \to 0^+} \frac{1}{2h} \int_{x-h}^{x+h} f_{k_0}(t)\,\mathrm{d}t = f_{k_0}(x) = f(x).$$

■

Referenciado en

- `teo-diferenciacion-lebesgue`

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